

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Caribbean ED Validation of Machine-Learning Triage Support

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Statement of the Problem:

Most AI-based triage tools have been developed and tested outside Caribbean emergency department settings, making local safety, fairness, and effectiveness uncertain. (1–10) At Mercer General Hospital, differences in patient populations, staffing, resources, diagnostic delays, ward-bed availability, clinical judgement, and existing workflows may affect AI-supported triage. (11–13) Therefore, this project addresses whether a machine-learning triage support tool can be locally validated and integrated alongside existing triage practice to improve high-risk patient identification while avoiding increased workload, alert fatigue, under-triage, over-triage, documentation burden, or patient safety risks within a realistic Caribbean ED context. (2,7–10,12,14–19)

Previous Work:

Tyler et al. and Da’Costa et al. reviewed the use of artificial intelligence and machine learning in emergency department triage. Both studies found that AI has potential to support patient prioritisation, risk prediction, resource allocation, and patient flow management. They also highlighted challenges including algorithmic bias, limited dataset diversity, clinician trust, accountability concerns, and the need for further validation before widespread implementation. Together, these reviews suggest that AI may provide useful decision support, but successful deployment requires careful evaluation within local clinical workflows.(1,2)

Araouchi and Adda, Kang et al., and Bavali-Gazik et al. developed machine-learning models to support emergency department triage and patient prioritisation. Across these studies, models such as stacking approaches, gradient boosting methods, and fusion models achieved strong predictive performance using routinely collected triage data. Despite these promising results, all three studies relied on data from individual institutions or specific patient populations, limiting confidence in how well the models would generalise to other healthcare environments. These findings reinforce the need for local validation before implementation in Caribbean emergency departments.(3,5,6)

Alomari et al. and Wang et al. evaluated the use of large language models for emergency triage and patient guidance. Both studies reported encouraging performance under controlled conditions, with GPT-based systems demonstrating strong agreement with existing triage approaches. However, limitations included reliance on structured scenarios, dataset-specific conditions, and reduced performance when compared with experienced clinicians. These findings suggest that large language models may serve as decision-support tools but should not replace clinical judgement.(4,14)

De Freitas et al. explored patient flow in a Caribbean emergency department using qualitative methods including observation, process mapping, and informal interviews. The study found that patient flow was influenced by work processes, emergency department layout, material resources, staffing levels, non-clinical personnel, and links with other hospital departments. Although the study did not focus on AI, it provides important regional context by demonstrating how local

workflow characteristics influence emergency department performance. This highlights why any AI triage support system should be validated within the healthcare environment where it will be used.(11)

Farrohknia et al. conducted a systematic review of emergency department triage scales, including the Emergency Severity Index (ESI), Canadian Triage and Acuity Scale (CTAS), Manchester Triage System (MTS), and Australasian Triage Scale (ATS). The review found that five-level triage systems generally demonstrated better reliability and validity than older three-level systems. However, the authors also noted that evidence supporting some triage components remained limited and varied across healthcare settings. This study is useful because it highlights the importance of understanding the strengths and limitations of existing triage frameworks before introducing AI-supported decision tools.(15)

Christ et al. and Tanabe et al. examined modern emergency department triage systems and the use of the Emergency Severity Index. Their findings showed that structured five-level triage systems can support patient prioritisation and help predict resource utilisation within emergency departments. Tanabe et al. further demonstrated that ESI scores were associated with resource consumption and length of stay, supporting their operational value. Together, these studies provide evidence that existing triage systems offer a useful foundation upon which AI-supported decision tools can be built.(16,17)

Banerjea and Carter investigated waiting and interaction times in a developing-country emergency department in Barbados. They identified significant delays associated with triage, physician assessment, laboratory investigations, and admission processes, demonstrating that patient flow is affected by multiple operational bottlenecks. The study provides Caribbean evidence that emergency department performance depends on more than clinical decision-making alone. This is relevant because any AI triage support system must operate within these existing workflow constraints.(13)

Cho et al. examined emergency department triage decision-making and the role of clinical information in patient prioritisation. The study highlighted that triage decisions are influenced by patient presentation, available information, clinician judgement, and the need for reassessment as patient conditions change. This is relevant to the proposed project because an AI-supported triage tool would need to fit within these existing decision-making processes rather than replace them. It also supports the need to consider workflow integration, staff usability, and patient safety when introducing decision-support tools into emergency department triage.(12)

Abu Arra et al. examined factors influencing nurses' clinical decision-making in emergency departments, while Johansson et al. investigated alarm fatigue among emergency department staff. Together, these studies highlight the influence of workload, organisational factors, information availability, alarm burden, and cognitive demands on clinical performance. The findings suggest that emergency clinicians already operate in information-rich and time-sensitive environments where excessive alerts may reduce effectiveness. These considerations are important because AI-supported triage systems should enhance decision-making without increasing cognitive workload or contributing to alert fatigue.(18,19)

Wong et al. and Wallace et al. provide important safety evidence for healthcare AI and digital triage tools. Wong et al. externally validated the Epic Sepsis Model, a widely implemented proprietary sepsis prediction model, and found poor discrimination and calibration, raising concerns about using clinical prediction tools without strong local validation. Wallace et al. reviewed digital and online symptom checker tools and found low diagnostic accuracy and variable triage accuracy, with the authors warning that reliance on these tools could create patient safety hazards. These studies are relevant because they show that AI or digital triage tools may appear useful in theory

but can still create risks of missed high-risk patients, excessive alerts, unsafe urgency advice, and poor generalization when used in real clinical settings. (8,9)

The World Health Organization and Smallman provide broader ethical and governance guidance for healthcare AI. WHO identifies key principles including human autonomy, safety, transparency, accountability, inclusiveness, equity, and sustainability. Smallman argues that healthcare AI should not only be evaluated at the level of the individual patient, but also by its effects on groups, communities, institutions, health systems, and future use. Together, these sources support the need to treat the proposed Mercer triage tool as a socio-technical system rather than only a prediction model. This means the project must consider not only accuracy, but also staff workload, patient safety, accountability, fairness, local workflow fit, and long-term sustainability. (7,10)

Gaps Identified:

The first gap is limited local validation, safety testing, and generalizability of AI-supported triage tools within Caribbean emergency department settings. Several AI triage studies reported strong performance, but many were developed using data from single hospitals, specific countries, structured scenarios, or narrow patient groups. Real-world safety evidence also shows that clinical prediction and digital triage tools may perform poorly or inconsistently when tested outside their original setting. For example, the Epic Sepsis Model raised concerns about weak external performance, while digital symptom checker studies showed low diagnostic accuracy and variable triage accuracy. This means Mercer General Hospital cannot assume that a model developed elsewhere would perform safely, fairly, or accurately for Caribbean patients, local staffing patterns, resource availability, diagnostic delays, ward-bed availability, and emergency department workflows. Caribbean patient-flow studies further show that local delays, staffing constraints, diagnostic bottlenecks, and ward-bed availability can strongly affect how any new triage support tool would work in practice. (3–6,8,9,11,13)

The second gap is limited evidence on practical workflow integration, human oversight, and ethical governance. Many AI triage studies focus mainly on model accuracy, while fewer explain how the tool would fit into existing triage systems such as ESI or how it would affect nurses, doctors, documentation burden, alert fatigue, accountability, patient trust, and patient flow. This is important because triage decisions are shaped by clinical judgement, workload, available information, resource needs, and reassessment during waiting. WHO guidance also highlights that healthcare AI should protect human autonomy, promote safety, support transparency, foster accountability, ensure equity, and remain sustainable. Smallman's multiscale ethics approach also supports evaluating healthcare AI beyond the individual patient level by considering its effects on staff, communities, institutions, and future use. For Mercer General Hospital, the proposed tool should therefore be tested as a supervised support layer beside existing triage methods, with attention to high-risk patient identification, under-triage, over-triage, staff usability, cognitive workload, workflow fit, fairness, accountability, and long-term sustainability. (1,2,7–10,12,14–19)

Proposed Solution:

To address the limited local validation of AI triage tools in Caribbean emergency department settings, this project proposes a 12-week pilot study to test a machine-learning triage support tool using de-identified emergency department data from Mercer General Hospital. The tool would not replace nurses, doctors, or existing triage systems. Instead, it would act as a clinically supervised decision-support layer beside current triage practice, helping staff identify patients who may be at higher risk while keeping the final triage decision under human clinical judgement. This approach is important because real-world AI safety concerns, including the Epic Sepsis Model and digital symptom checker studies, show that clinical prediction and triage-support tools require careful validation before they are trusted in practice. (1,2,8,9,11–13)

The pilot would use routinely collected triage information such as age, sex, vital signs, chief complaint, mode of arrival if available, and existing triage category. A simple baseline model, such as logistic regression or a decision tree, would be developed first because these models are easier to interpret, explain, and compare with current triage decisions. If time allows, the baseline model could then be compared with stronger machine-learning models such as Random Forest or Gradient Boosting to assess whether more complex models improve performance enough to justify their use in a clinical support setting. This supports the need for explainability, transparency, and safe implementation rather than assuming that a more complex model is automatically better. (3,5–7,10)

The model would be evaluated using clinically relevant safety measures rather than accuracy alone. These would include high-acuity recall, under-triage risk, over-triage rate, false-negative rate, alert burden, agreement with existing triage outcomes, and the model's ability to identify patients who may need urgent review, admission, or higher-level care. Where data are available, performance should also be reviewed across patient subgroups to check whether the tool performs differently for groups such as elderly patients, paediatric patients, pregnant patients, rural patients, or walk-in patients. This is important because an AI triage tool must reduce the risk of missing high-risk patients while also avoiding unnecessary pressure on limited emergency department resources. (1,2,4,7–9,14)

To address the workflow integration and human-factor gap, a simple workflow mock-up would also be created to show how the tool could fit into Mercer General Hospital's existing triage process. The mock-up would show where a risk score or alert could appear, how it could sit beside an existing triage framework such as ESI, and how nurses and doctors might interpret the output without adding unnecessary documentation work. The design would also consider clinical reassessment, alert fatigue, cognitive workload, staff usability, patient flow during waiting periods, and the ability of staff to override the AI recommendation when clinically appropriate. This is important because Caribbean emergency department performance is influenced by local staffing, diagnostic delays, physical layout, material resources, ward-bed availability, and links with other hospital departments. (11–13,15–19)

The pilot would remain within a safe 12-week scope. It would use only de-identified data and would not involve live hospital deployment, autonomous decision-making, regulatory approval, or direct changes to patient care. Any future clinical use would require stronger validation, clinical oversight, stakeholder feedback, subgroup performance review, clear accountability, and hospital or ethics approval. Based on WHO guidance and multiscale ethics considerations, the proposed system should be treated as a socio-technical intervention, meaning its success would depend not only on model performance, but also on safety, fairness, transparency, staff trust, patient protection, and long-term sustainability within Mercer General Hospital. (7,10)

Workflow and AI Plug-in Points

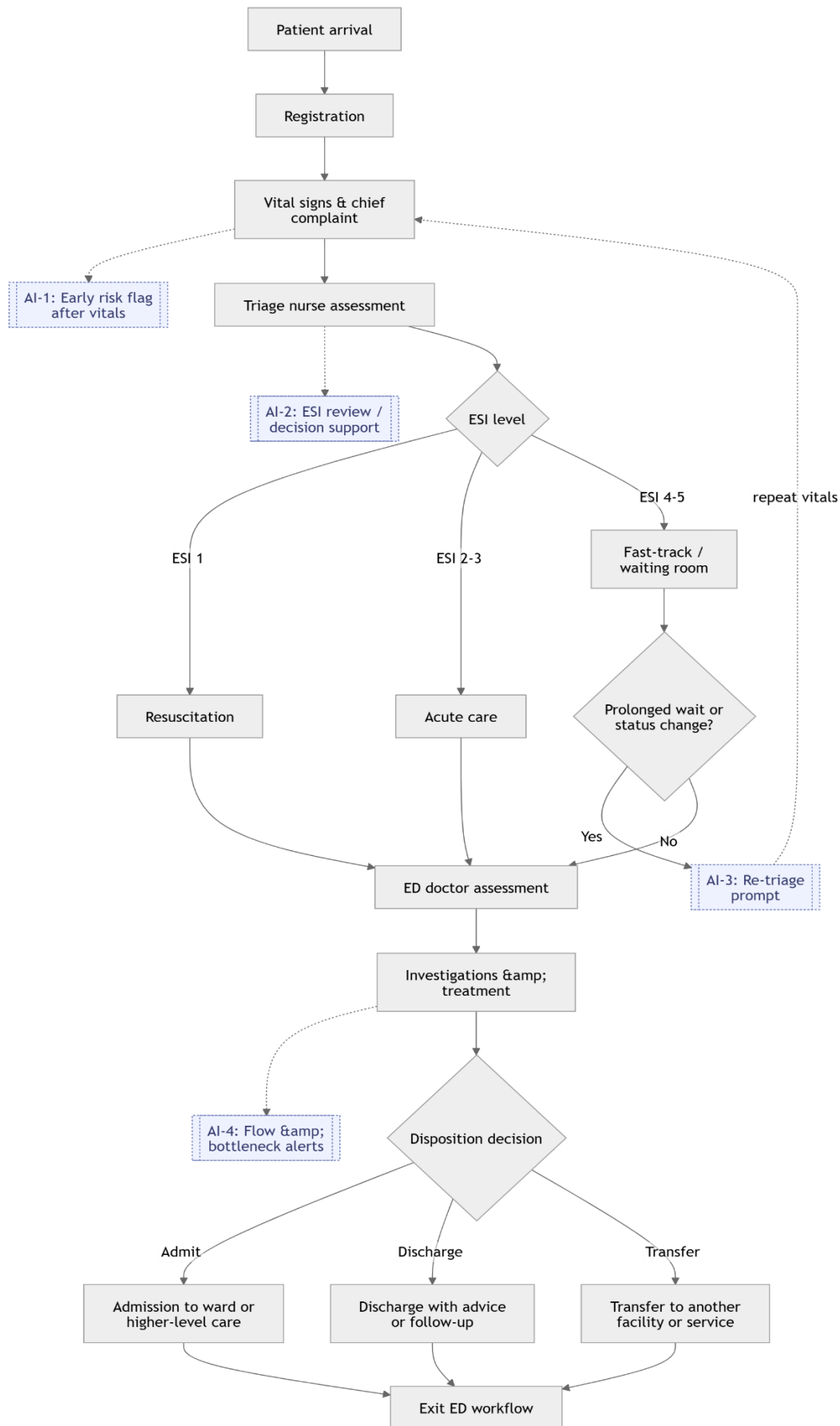


Figure 1: Proposed workflow position of a machine-learning triage support tool within the Mercer General Hospital emergency department triage process.

The proposed system would fit within the existing Mercer General Hospital emergency department workflow rather than replacing the current triage process. As shown in Figure 1, the workflow begins with patient arrival by ambulance, walk-in, private vehicle, taxi, or EMS. This is followed by registration, vital-sign collection, triage nurse assessment, Emergency Severity Index (ESI) assignment, care-area allocation, ED doctor assessment, investigations or treatment, disposition, and exit from the emergency department.

The first AI plug-in point occurs after vital-sign collection. At this stage, the system could use information already available during triage, such as age, sex, arrival mode, chief complaint, vital signs, pain score, allergies, and brief clinical history. The output would be an early risk flag for abnormal vital-sign patterns or possible high-risk presentation. This would support the triage nurse's assessment while keeping the final decision under clinical control.

The second AI plug-in point occurs after the ESI level is assigned. At this stage, the system could compare the assigned ESI category with the model's estimated risk level to identify possible mismatch. For example, it could flag patients who may have been under-triaged or over-triaged based on their recorded observations and presenting features. This would allow the nurse or doctor to review the case without replacing the existing ESI framework.

The third AI plug-in point occurs during waiting-room or fast-track monitoring. Since triage is not a one-time event, patients may deteriorate while waiting, especially during periods of high patient volume or delayed assessment. The system could monitor waiting time, initial ESI level, and previously recorded observations to prompt repeat vital signs or reassessment when appropriate. This would support re-triage and help reduce the risk of missing patients whose condition changes after initial triage.

The fourth AI plug-in point occurs during investigations, treatment, and patient-flow monitoring. At this stage, the system could highlight delays linked to laboratory investigations, imaging, specialist consultation, or ward-bed availability. This function would not make clinical decisions independently but could support charge nurses and doctors by improving visibility of bottlenecks that affect patient movement through the department.

Overall, the workflow diagram positions the proposed AI system as a low-friction decision-support layer within existing clinical practice. Its role would be to support early risk recognition, ESI review, waiting-room reassessment, and patient-flow awareness while maintaining clinician oversight, avoiding unnecessary documentation burden, and respecting the operational realities of a Caribbean emergency department.

GitHub Repository: <https://github.com/Kari-2005/carisurg-portfolio>

The Week 3 workflow diagram source file and exported diagram are available in the repository under docs/Week3/.

Constraints & Stakeholders:

Three main workflow constraints must be considered before introducing an AI-supported triage tool at Mercer General Hospital.

The first constraint is paper-first documentation and delayed digital data availability. Patient information and vital signs may initially be recorded on paper before being entered into an electronic system. This creates a delay between when information is available to clinicians and when it becomes available digitally. An AI system that relies entirely on real-time electronic health record data may therefore be unreliable at the point where triage decisions are being made. To

reduce this risk, the proposed tool should use information already collected during triage and support gradual structured digital data capture.

The second constraint is limited triage nurse time and cognitive workload. Triage nurses work under significant time pressure, especially during periods of high patient volume. Additional data entry, complex interfaces, or excessive alerts could increase workload and disrupt existing triage practice. If the tool slows staff down or produces too many alerts, clinicians may ignore it or avoid using it. Therefore, the system should function as low-friction decision support, rely mainly on existing triage information, and present concise, clinically relevant outputs.

The third constraint is patient deterioration during waiting and the need for re-triage. Patient conditions may change after the initial triage decision, particularly during prolonged waits. A model that only supports the first ESI assignment may miss deterioration later in the patient journey. The proposed system should therefore support repeat observations and generate prompts for reassessment when appropriate, while keeping final decisions under clinical supervision.

Table 1: Workflow constraints, design impact, and mitigation strategies for the proposed AI-supported triage workflow.

Workflow constraint	Design impact	Mitigation
Paper-first documentation and delayed digital data availability	The model may not access triage information at the exact point when decisions are being made.	Use information already collected during triage and support gradual structured digital data capture.
Limited triage nurse time and cognitive workload	Extra data entry, complex screens, or excessive alerts may cause staff to ignore or avoid the tool.	Keep the tool low-friction, concise, and based mainly on existing triage information.
Patient deterioration during waiting and need for re-triage	A model based only on initial triage may miss patients whose condition worsens while waiting.	Include prompts for repeat observations and reassessment when appropriate, with final decisions kept under clinical supervision.

Table 2: Stakeholder map for integrating a machine-learning triage support tool into Mercer General Hospital's emergency department workflow.

Stakeholder	Main concern	What success looks like
Triage nurse	The tool must not slow triage, increase documentation burden, or undermine clinical judgement.	The nurse receives a clear risk flag using information already collected during triage and can make the final decision quickly.
ED doctor / consultant	The tool must provide reliable and clinically defensible risk information.	High-risk patients are easier to identify for urgent review without replacing clinical assessment.
Nurse-in-charge / charge nurse	The tool must improve visibility of patient flow without adding confusion during busy shifts.	Waiting-room risk, bottlenecks, and staff workload are easier to monitor and act on.
Registration / administrative staff	The tool must not delay intake or create duplicate data-entry work.	Patient information is captured accurately once and can support triage without extra administrative burden.
Patient advocate / patient representative	The tool must protect patient safety, fairness, privacy, and timely care.	Patients are prioritised fairly, personal data are protected, and high-risk patients are not missed.

Risk Analysis:

The proposed machine-learning triage support tool carries safety, operational, ethical, and equity risks because it would influence emergency department prioritisation, even though it would not make the final triage decision. Real-world AI safety evidence shows why this matters. Wong et al. found that the Epic Sepsis Model performed poorly when externally validated, while Wallace et al. found that digital and online symptom checker tools had low diagnostic accuracy and variable triage accuracy. These findings show that clinical prediction and digital triage tools can create patient-safety risks if they are trusted without local validation.(8,9) WHO guidance also emphasizes that healthcare AI should protect human autonomy, promote safety, support transparency, foster accountability, ensure equity, and remain sustainable.(7) Therefore, the proposed Mercer triage tool should be treated as a supervised socio-technical support system rather than only a prediction model. (7,10)

For this proposal, likelihood and impact are rated as Low, Medium, or High. Likelihood refers to how likely the risk is to occur during development, testing, or future use. Impact refers to the possible effect on patient safety, staff workload, fairness, trust, or emergency department workflow.

Table 3: Risk register for the proposed machine-learning triage support tool.

Risk name	Category	Likelihood	Impact	Mitigation	Signal of success
Distribution shift	AI-technical	Medium	High	Validate the model using local Mercer General Hospital data before any future use. Recheck performance over time, especially during outbreaks, seasonal changes, staffing shortages, or workflow changes.	High-acuity recall remains stable over time, and any major performance drop triggers clinical review.
Single-site overfitting	AI-technical	High	High	Start with simple baseline models and avoid assuming that performance at Mercer General Hospital will generalise to other Caribbean hospitals without external validation.	Model performance on holdout data remains close to training performance, and wider rollout is paused until external validation is completed.
Data leakage	AI-technical	Medium	High	Audit all model inputs to confirm that only information available at or before triage is used. Remove post-triage variables such as final diagnosis, admission outcome, length of	Feature review confirms that all model inputs would realistically be available during triage.

				stay, or discharge status from training inputs.	
Poor explainability	AI-technical	Medium	Medium	Prefer interpretable models such as logistic regression or decision trees during the pilot. Provide short explanations for risk flags where possible.	Triage staff can understand the main reason for a risk flag in plain clinical language during testing.
Alert fatigue	Operational	High	High	Limit alerts to clinically important high-risk cases. Review alert frequency with triage staff and adjust thresholds if too many alerts are ignored or dismissed.	A reasonable proportion of alerts leads to clinical review or action, with no steady increase in ignored alerts.
Workflow disruption	Operational	Medium	High	Design the tool as a low-friction support layer using information already collected during triage. Avoid duplicate data entry, complex screens, or long explanations during busy shifts.	Triage time does not increase meaningfully during workflow testing, and staff report that the tool is usable.
System downtime or delayed data entry	Operational	Medium	Medium	Maintain a paper-based and clinician-led fallback process. The AI tool should not be required for triage to continue safely.	Staff can complete triage safely during downtime or delayed digital entry without relying on the AI tool.
Automation bias	Ethical	High	High	Keep the tool as decision support only, with final decisions made by nurses and doctors. Train staff to question, override, and document disagreement with AI recommendations.	Staff correctly override unsafe AI recommendations during training or simulation cases.
Accountability ambiguity	Ethical	Medium	High	Define who is responsible for reviewing model errors, monitoring performance, and responding to safety concerns before any future deployment.	A written accountability process names the clinical owner and explains how AI-related incidents would be reviewed.

Consent and disclosure gap	Ethical	Medium	Medium	Develop simple patient-facing wording explaining that AI may support triage but does not replace clinicians. Review this wording with hospital leadership and patient representatives.	Patient-facing disclosure language is reviewed and approved before any live clinical use.
Underrepresentation of patient groups	Equity	High	High	Review model performance across available patient groups, such as age, sex, arrival mode, pregnancy status if available, rural patients, walk-in patients, and patients with multiple conditions.	False-negative rate and high-acuity recall are reviewed by subgroup, and deployment is paused if unsafe differences are found.
Language and health-literacy barriers	Equity	Medium	Medium	Use plain, non-technical wording in staff-facing alerts and patient-facing explanations. Avoid wording that could confuse staff or patients.	Staff and patient representatives confirm that the wording is clear and understandable.

The three highest-priority risks are automation bias, distribution shift, and underrepresentation of patient groups. Automation bias is high priority because emergency departments are busy, time-sensitive environments, and staff may over-trust a risk score when they are under pressure. If the model gives an incorrect recommendation and staff follow it without question, a high-risk patient could be under-triaged. This risk would be reduced by keeping the tool supervised, allowing clinicians to override, and the training of staff with simulation cases where the AI recommendation is intentionally incorrect. Distribution shift is also high priority because the tool may perform differently when patient patterns, resources, or workflows change. A model trained on one set of Mercer General Hospital data may not work the same way during an outbreak, mass-casualty event, staffing shortage, or future deployment in another Caribbean hospital. This risk would be reduced through local validation, holdout testing, and repeated performance checks over time. Underrepresentation of patient groups is the third high-priority risk because a model can appear accurate overall while performing worse for specific groups. Elderly patients, paediatric patients, pregnant patients, rural patients, walk-in patients, or patients with multiple conditions may be less well represented in the training data. If the model performs worse for these groups, it could worsen existing healthcare inequalities. This risk would be reduced by reviewing performance across available subgroups and pausing future deployment if unsafe differences are found.

Overall, the risk analysis shows that the proposed tool should not be deployed as an autonomous triage system. Its safest role is as a supervised decision-support layer that helps clinicians identify possible high-risk patients while preserving human judgement, local workflow fit, accountability, fairness, and patient safety. Within the 12-week pilot, the focus should remain on retrospective validation, workflow design, ethics review, and risk monitoring rather than live clinical use.

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